UNIVERSITY OF CAPE COAST
COLLEGE OF AGRICULTURE AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF PHYSICS
FIRST SEMESTER: 2022/2023
PHY 101: GENERAL PHYSICS I – GROUP 15
JAN, 2023
LECTURE ONE WORK EXERCISES VERSION 2

1. What is a physical quantity?

2. Differentiate between fundamental and derived quantities.

3. Define Dimension.

4. Write the dimensional formula of the following physical quantities

(i) Potential Energy (ii) Power (iii) Surface Tension (iv) Strain(v) stress

5. Check the correctness of the relation λ = h /mv; where λ is wavelength, h- Planck's constant, m is mass of the particle and v - velocity of the particle.

6. Check whether the following equations are dimensionally correct.

(i) v = u + at

(ii) $F = \frac{MV}{r^2}$

(iii) $v^2 - u^2 = 2as$

7. What is the formula, S.I units and physical quantity of the following

(a) $[ML^{-1}T^{-2}]$

(b) $[M^1L^2T^{-2}]$

(c) $[M^1L^1T^{-1}]$

8. The frequency *f* of a transverse wave in a stretched string depends on the length L, tension T, and linear density *m*. Deduce the formula for f in terms of L, T and m using dimensional analysis.

9. How many significant figures do we have in the following?

i. 4250
ii. 0.00703
iii. 5030.0
iv. 750.064080

10. Express the speed of light $3.0\times10^8 ms^{-1}$ in millimeters per picosecond (mm/ps).

11. Consider two vectors **A** = 3**i** – 2**j** and **B** = – **i** – 4**j**. Calculate:

i. $\left(\vec{A}+\vec{B}\right)$ and $\left(\vec{A}-\vec{B}\right)$.
ii. |A + B| and |A – B|
iii. The direction of $\left(\vec{A}+\vec{B}\right)$ and $\left(\vec{A}-\vec{B}\right)$.

12. A force of 3N acts on 90° to a force of 4N. Find the magnitude and direction of their resultant, R.

13. Consider two vectors **A** = 6**i** – 2**j** +4**k** and **B** = – **i** – 4**j**+2**k**. Calculate:

i. $\left(\vec{A}+\vec{B}\right)$ and $\left(\vec{A}-\vec{B}\right)$.

ii. the angle between $\underline{A}$ and $\underline{B}$.

iii. the unit vector of $\left(\vec{A}+\vec{B}\right)$ and $\left(\vec{A}-\vec{B}\right)$.

iv. $\left|\vec{A}+\vec{B}\right|$ and $\left|\vec{A}-\vec{B}\right|$.

14. A car travels 45km due north and then 60km in a direction 75^0 west of north. Find the magnitude and direction of the car's resultant displacement.